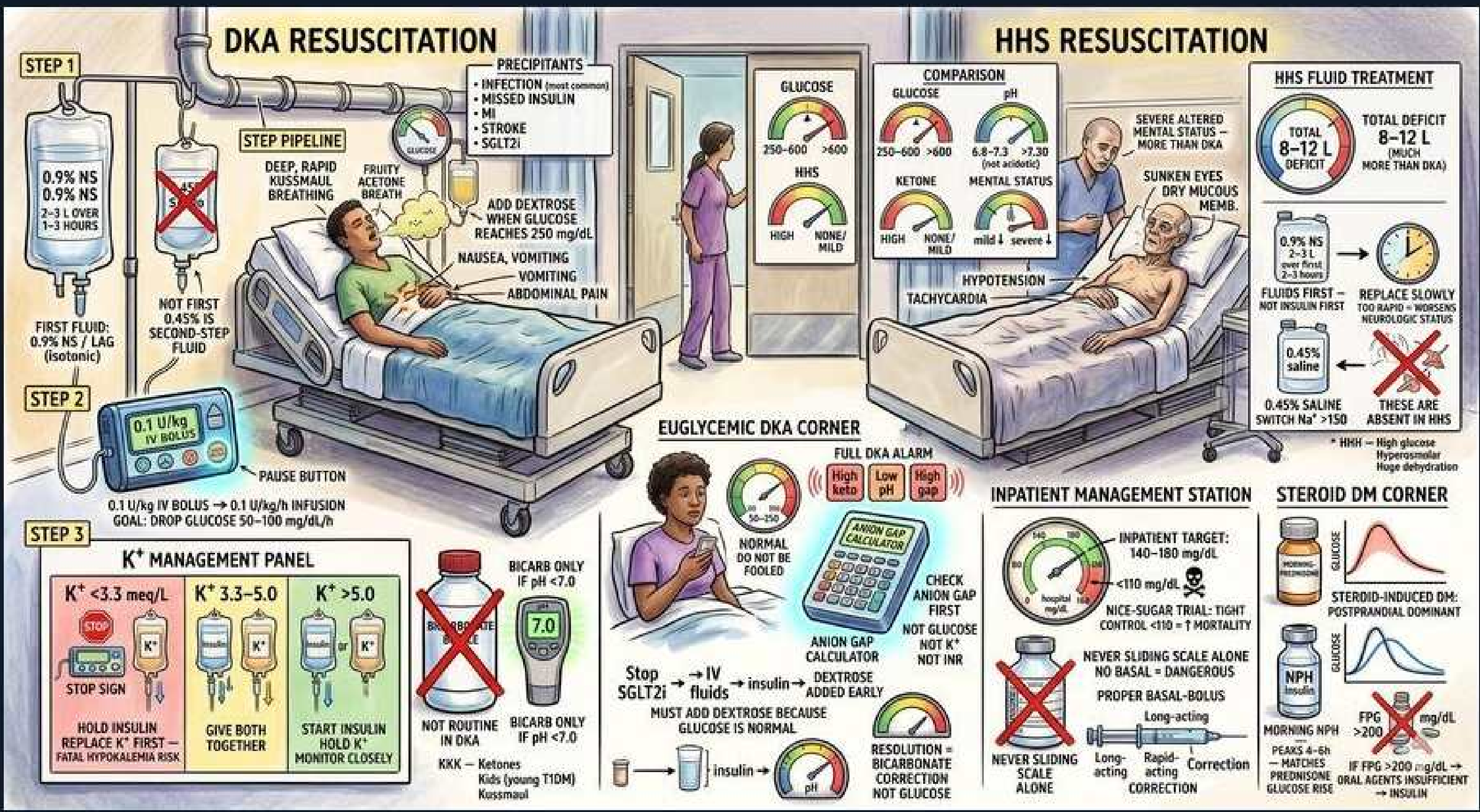
DM — Scene 2 — Acute Hyperglycemic Treatment: DKA/HHS Management



1. Step 1 Fluids first

WHAT YOU SEE

Far-LEFT STEP 1: first IV bag, 0.9% NS / LR isotonic, 2-1 L over 1-3 hours (first fluid)

WHAT IT ENCODES

Rung 1 (always first): isotonic IV fluids — 0.9% NS (or LR) at ~15-20 mL/kg or 1-1.5 L in the first hour. Deficits: DKA ~3-6 L, HHS ~8-12 L (more profound dehydration → more aggressive resuscitation). Fluids alone lower glucose substantially via hemodilution and restored renal perfusion/clearance, and they restore hemodynamics before insulin shifts fluid/K+ intracellularly.

BOARD TRAP

WRONG: 'Start the insulin drip immediately on arrival.' Insulin before volume resuscitation worsens hypovolemia and can cause fatal hypokalemia. Discriminator: fluids are rung 1 — insulin waits for volume + a checked K+.

---

2. Correct Na -> switch to half-NS

WHAT YOU SEE

Far-LEFT: second IV bag 0.45% NS = NOT FIRST, second-step fluid after Na correction

WHAT IT ENCODES

After the initial NS bolus, choose maintenance fluid by the CORRECTED sodium (add ~1.6-2.4 mEq Na per 100 mg/dL glucose above normal, since hyperglycemia dilutes measured Na). If corrected Na is normal/high → switch to 0.45% (half) NS to give free water; if corrected Na is low → continue 0.9% NS. Goal: replace the free-water deficit while avoiding too-rapid osmolar shifts.

BOARD TRAP

WRONG: 'The measured serum Na is low, so give hypotonic fluids and treat hyponatremia.' The low measured Na is usually a dilutional artifact of hyperglycemia — correct it first. Discriminator: act on the CORRECTED sodium, not the raw value.

---

3. Step 2 Potassium before insulin

WHAT YOU SEE

Bottom-LEFT STEP 3 header: K+ MANAGEMENT PANEL (potassium checked/managed before insulin)

WHAT IT ENCODES

Rung 2: check serum potassium BEFORE starting insulin, because insulin (and correction of acidosis) drives K+ into cells and can precipitate life-threatening hypokalemia/arrhythmia. Total-body K+ is depleted in every DKA/HHS patient regardless of the serum value, due to osmotic-diuresis losses and transcellular shift. Recheck K+ q2h during treatment.

BOARD TRAP

WRONG: 'Glucose is dangerously high, so start insulin first and check potassium afterward.' Insulin given before knowing K+ can cause fatal hypokalemia. Discriminator: K+ is checked and managed BEFORE the insulin drip — always.

---

4. K traffic light <3.3 hold insulin

WHAT YOU SEE

RED light K bag, K+ <3.3 mEq/L: STOP SIGN, HOLD INSULIN, replace K first - fatal hypokalemia risk

WHAT IT ENCODES

RED light — K+ <3.3 mEq/L: HOLD insulin and replete potassium first (give ~10-20 mEq/hr IV) until K+ >=3.3. Insulin here would shift the already-low K+ intracellularly and cause fatal hypokalemia/arrhythmia. Resume insulin only once K+ has crossed back above 3.3.

BOARD TRAP

WRONG: 'K+ is 3.0 but glucose is 500 — start insulin anyway, the hyperglycemia is the emergency.' Hypokalemia is the more immediate killer (arrhythmia). Discriminator: K+ <3.3 → replete potassium FIRST, hold insulin, no exceptions.

---

5. K 3.3-5.2 give K with fluids

WHAT YOU SEE

YELLOW light K bag, K+ 3.3-5.0 mEq/L: GIVE K WITH FLUIDS / insulin together

WHAT IT ENCODES

YELLOW light — K+ 3.3-5.2 mEq/L: proceed with insulin AND add 20-30 mEq potassium to each liter of IV fluid to keep K+ in the 4-5 range, anticipating the fall as insulin acts. This is the most common scenario at presentation. Continue cardiac monitoring and q2h K+ checks.

BOARD TRAP

WRONG: 'K+ is normal at 4.5, so no potassium replacement is needed.' A 'normal' serum K+ in DKA still means total-body depletion and will drop with insulin. Discriminator: in the 3.3-5.2 window, give K+ WITH the insulin — a normal value is not reassurance.

---

6. K >5.2 no K yet

WHAT YOU SEE

GREEN light K bag, K+ >5.0 mEq/L: NO K yet, start insulin, recheck

WHAT IT ENCODES

GREEN light — K+ >5.2 mEq/L: start insulin and fluids but do NOT add potassium yet; recheck K+ in ~2 hours. Insulin and rehydration will bring the elevated serum K+ down (and total-body K+ is still low, so it will eventually need replacement). Begin K+ once it falls below 5.2 and urine output is confirmed.

BOARD TRAP

WRONG: 'K+ is 5.5 — treat the hyperkalemia with kayexalate/aggressive measures.' The hyperkalemia is from transcellular shift and self-corrects with insulin/fluids. Discriminator: hold K+ replacement (don't aggressively lower it either) and recheck — insulin itself is the treatment.

---

7. Step 3 Insulin infusion

WHAT YOU SEE

Far-LEFT STEP 2: insulin syringe/pump, 0.1 U/kg/h IV infusion, goal drop 50-100 mg/dL/h

WHAT IT ENCODES

Rung 3: IV regular insulin infusion at 0.1 U/kg/hr (with or without an initial 0.1 U/kg bolus; the bolus is OMITTED in children to reduce cerebral-edema risk). Target a glucose fall of 50-100 mg/dL/hr; if glucose doesn't drop ~10% in the first hour, increase the rate or reassess fluids. Insulin's real job is to halt lipolysis/ketogenesis and close the anion gap.

BOARD TRAP

WRONG: 'Use intermittent subcutaneous regular insulin boluses for severe DKA.' Variable SC absorption in a dehydrated patient is unreliable — IV infusion is standard for severe DKA/HHS. Discriminator: continuous IV insulin (not SC) until the gap closes.

---

8. Step 4 Add dextrose at ~200

WHAT YOU SEE

Center-bottom: ADD DEXTROSE when glucose reaches ~200 (euglycemic-DKA corner), continue insulin

WHAT IT ENCODES

Rung 4: when glucose reaches ~200 mg/dL (DKA) or ~250-300 (HHS), add D5 (e.g., D5-half-NS) to the fluids and CONTINUE the insulin infusion. The dextrose lets you keep insulin running to clear ketones/close the gap without causing hypoglycemia. You may reduce the insulin rate slightly but do not stop it.

BOARD TRAP

WRONG: 'Glucose dropped to 200, so discontinue the insulin drip.' The anion gap is usually still open; stopping insulin reopens ketogenesis. Discriminator: at glucose ~200 → ADD dextrose, KEEP insulin — resolution is defined by the gap, not the glucose.

---

9. Step 5 Bicarb not routine

WHAT YOU SEE

Center-bottom: crossed-out bicarb bag, NOT ROUTINE IN DKA

WHAT IT ENCODES

Rung 5: bicarbonate is NOT routine — insulin + fluids correct the acidosis by clearing ketoacids (each metabolized ketone regenerates bicarbonate). Reserve it for severe acidemia, pH <7.0 (some use <6.9), where cardiac contractility and pressor responsiveness are impaired. Give as a slow infusion, never a rapid push.

BOARD TRAP

WRONG: 'pH is 7.1 — give bicarbonate to speed correction.' No benefit at pH >7.0, and it adds harm. Discriminator: the ONLY bicarb indication in DKA is pH <7.0; moderate acidosis resolves with insulin and fluids alone.

---

10. Bicarb risks

WHAT YOU SEE

Center-bottom: bicarb vial labeled pH 7.0, BICARB ONLY pH <7.0

WHAT IT ENCODES

Why bicarbonate is avoided: it worsens HYPOKALEMIA (drives K+ intracellularly), causes a PARADOXICAL CSF/CNS acidosis (CO2 generated crosses the blood-brain barrier faster than bicarbonate), promotes arrhythmias, delays ketone clearance, can cause rebound metabolic alkalosis, and increases cerebral-edema risk in children.

BOARD TRAP

WRONG: 'Bicarbonate is harmless and at worst just doesn't help.' It actively causes hypokalemia, paradoxical CNS acidosis, and pediatric cerebral edema. Discriminator: bicarb is potentially harmful, which is precisely why it's withheld unless pH <7.0.

---

11. Step 6 Transition / anion gap closed

WHAT YOU SEE

Bottom-RIGHT corner: ANION GAP CALCULATOR, CHECK ANION GAP FIRST not glucose, RESOLUTION = correction not glucose

WHAT IT ENCODES

Rung 6: stop the IV insulin only after RESOLUTION — anion gap closed (and HCO3 >=15-18, pH >7.3, patient eating). DKA resolution criteria are gap/bicarb-based, NOT a normal glucose. Before stopping the drip, give long-acting SC (basal) insulin and OVERLAP it with the infusion for 1-2 hours so a basal level is established when the drip comes off.

BOARD TRAP

WRONG: 'The glucose normalized, so stop the IV insulin now.' Glucose normalizes long before the gap closes; stopping early causes rebound DKA. Discriminator: transition is triggered by gap closure (not glucose) and requires SC overlap before the drip stops.

---

12. Prevent rebound DKA

WHAT YOU SEE

Bottom-center transition: must add dextrose / continue coverage so glucose-normal does not stop insulin (prevent rebound)

WHAT IT ENCODES

IV regular insulin has a very short half-life (~minutes); if the drip is stopped before SC basal insulin has taken effect, insulin levels crater and ketoacidosis rebounds. Give SC basal (e.g., glargine) and overlap with the infusion for 1-2 hours (rapid-acting nutritional insulin overlaps ~15-30 min) so coverage is continuous.

BOARD TRAP

WRONG: 'Turn off the drip and give the first SC dose at the same moment.' The gap between drip-off and SC onset reopens ketogenesis. Discriminator: SC insulin must be given 1-2h BEFORE stopping the IV drip — no insulin-free gap.

---

13. Complications: cerebral edema

WHAT YOU SEE

RIGHT HHS fluid panel: REPLACE SLOWLY - TOO RAPID WORSENS NEUROLOGIC STATUS (over-rapid osmolar correction warning)

WHAT IT ENCODES

Cerebral edema is the most feared, leading cause of DKA death in CHILDREN, caused by over-rapid fluid/osmolar correction (rapid drop in glucose/osmolality shifts water into brain cells). Prevention: avoid the insulin bolus in kids, lower glucose no faster than ~50-100 mg/dL/hr, and correct osmolality gradually. Onset: headache, lethargy, bradycardia/HTN (Cushing), declining consciousness 4-12h into treatment → give mannitol/hypertonic saline.

BOARD TRAP

WRONG: 'In pediatric DKA, correct fluids and glucose as fast as possible to fix the acidosis quickly.' Aggressive/rapid correction precipitates fatal cerebral edema. Discriminator: in children, go SLOW (no bolus, controlled osmolar correction) — speed kills here.

---

14. Complications: hypoglycemia/hypoK/hypophos/HCMA

WHAT YOU SEE

RIGHT HHS panel electrolyte/fluid warnings: 0.45% saline switch Na>150, slow replacement, large-deficit complications

WHAT IT ENCODES

Treatment-related complications: HYPOGLYCEMIA (from continuing insulin — prevent by adding dextrose at glucose ~200); HYPOKALEMIA (insulin/bicarb shift K+ in — replace proactively); HYPOPHOSPHATEMIA (insulin drives phosphate intracellularly — replace only if severe <1.0 mg/dL or with cardiac/respiratory/muscle weakness); HYPERCHLOREMIC NON-GAP ACIDOSIS (from large-volume NS resuscitation — expected, benign, self-resolves).

BOARD TRAP

WRONG: 'A persistent acidosis after the anion gap has closed means the DKA is not resolved — restart aggressive insulin.' A post-treatment NON-gap (hyperchloremic) acidosis from NS is benign and self-limited. Discriminator: check the gap — if it's closed, a residual non-gap acidosis is the saline artifact, not ongoing DKA.
